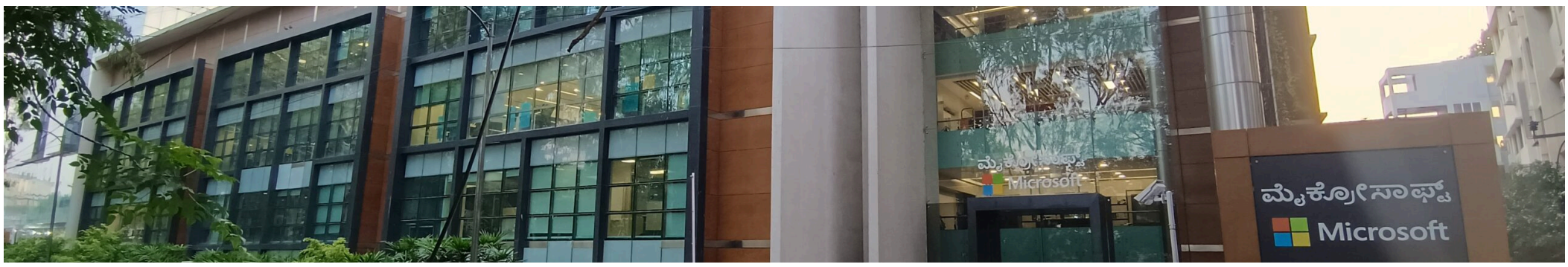

Microsoft Research Lab – India

Podcast: Helping young students build a career in research through the MSR India Research Fellow program. With Shruti Rijhwani and Dr. Vivek Seshadri

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One of Microsoft Research India's goals is to help strengthen the research ecosystem and encourage young students to look at research as a career. But it is not always easy for students to understand what research is all about and how to figure out if research is the right career for them. The Research Fellow program at Microsoft Research India enables bright young students to work on real-world research problems with top notch researchers across the research lifecycle, including ideation, implementation, evaluation, and deployment. Many of the students who have been part of the program have gone on to become researchers, engineers and entrepreneurs.

Today, we speak to Shruti Rijhwani, a graduate of MSR India's Research Fellow program who is currently doing her PhD at the Carnegie Mellon University, and joining us is Dr. Vivek Seshadri, a researcher at MSR India who also heads the Research Fellow program at the lab.

Shruti was a research fellow at MSR India in 2016, working on natural language processing models for code-switched text.

She is currently PhD student at the Language Technologies Institute at Carnegie Mellon University. Stemming from her work at MSR India, she has continued research in multilingual NLP, with a focus on low-resource and endangered languages.

Vivek primarily works with the Technology for Emerging Markets group at Microsoft Research India. He received his bachelor's degree in Computer Science from IIT Madras, and a Ph.D. in Computer Science from Carnegie Mellon University where he worked on problems related to Computer Architecture and Systems. After his Ph.D., Vivek decided to work on problems that directly impact people, particularly in developing economies like India. Vivek is also the Director for the Research Fellow program at MSR India.

For more information about the Research Fellow program, click here.

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Transcript

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[Music]

Sridhar Vedantham: Welcome to the Microsoft Research India podcast, where we explore cutting-edge research that's impacting technology and society. I'm your host, Sridhar Vedantham.

[Music]

One of Microsoft Research India's goals is to help strengthen the research ecosystem and encourage young students to look at research as a career. But it is not always easy for students to understand what research is all about and how to figure out if research is the right career for them. The Research Fellow program at Microsoft Research India enables bright young students to work on real-world research problems with top notch researchers across the research lifecycle, including ideation, implementation, evaluation, and deployment. Many of the students who have been part of the program have gone on to become researchers, engineers and entrepreneurs.

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[Music]

Sridhar Vedantham: OK, so I'm looking forward to this podcast because it's going to be a little different from what we've done in the past, in the sense that this is not a podcast about research projects or technologies, but it's something much more human, and we're going to be talking about the Research Fellow program that we have at MSR India.

And I'd like to welcome a special guest- Shruti, who used to be a Research Fellow at the lab and also Vivek Seshadri, who is a researcher at the lab and whom we've had on the podcast earlier in a different capacity. But today he's wearing the hat of the czar or the director of the Research Fellow program here.

So welcome, Shruti and Vivek.

Vivek Seshadri: Good evening, Sridhar, and very good morning Shruti.

Shruti Rijhwani: Hi Sridhar and Vivek, it's great to be here and great to be back to interacting with people from MSR India. It's been about four years since I left the RF program, so I'm really looking forward to talking about it and remembering some of my experiences.

Sridhar Vedantham: Excellent, so let's lay a little bit of a groundwork before we jump into the whole thing. Vivek can you give us a bit of an overview of what the Research Fellow program is.

Vivek Seshadri: Sridhar, the Research Fellow program has been around ever since the organization Microsoft Research India started itself. I think initially it was called the Assistant Researcher program and then it was called the Research Assistant Program and right now we're calling it the Research Fellow program. But the core of the program has been to enable recent undergraduate and Master's students to spend one or two years at MSR India and get a taste for what research in computer science looks like, especially in an industrial setting.

Sridhar Vedantham: So has the program evolved over time, or have there been any substantive changes or it's still the same at its core and its essence?

Vivek Seshadri: I think the only thing that has changed significantly is the number of research fellows that we have had. I think in the program started in its first year, I think we had had three assistant researchers, and today as we speak, we have over 50 Research Fellows in the lab working on various projects. So, in that sense, the program has definitely grown in size along with the lab, but I think the core goal of the program has not changed at all. It is still to give Research Fellows a taste of what research looks like in computer science and enable them to build their profile and prepare them for a career in computer science research and engineering.

Sridhar Vedantham: Right, so one thing that I've seen personally is that the Research Fellows add a huge amount of energy and life to the lab. And on that note, Shruti, what motivated you to come into MSR India to join the Research Fellow program?

Shruti Rijhwani: That's a great question and something that I think my experience is probably what a lot of Research Fellows who apply and join the program actually go through before deciding to join the program. So, I did an undergrad degree in computer science from BITS PILANI and during those four years I took classes on machine learning, information retrieval and so on, and also did two internships where I got a taste of like how machine learning applications can be applied to products in the real world. But both of those internships were kind of focused on the engineering side and I was really interested in what a career in doing machine learning research or using machine learning for research-based applications would look like. And I knew that if I wanted to pursue a career in this field, I would probably have to go to graduate school to get a Master's and a PhD, but I wasn't entirely sure whether this is what I wanted to do, so it was kind of like an exploratory phase for me to be in the Research Fellow program. I wanted to get some research experience, I wanted to see what established researchers in computer science do on a daily basis, and what the research process kind of is like when you're working in machine learning, and more specifically in natural language processing, which is what I was interested in.

Sridhar Vedantham: Right, so, Vivek, what I'm getting from Shruti is that, uh, the Research Fellow program is something that she is looking at to form a base or a basis for a longer career in research itself. So do you have any specific insights or inputs into about what the program actually offers in a structured manner to the Research Fellows that we have?

Vivek Seshadri: Yeah, so Microsoft Research now at its core is a research organization. All researchers at MSR do research in computer science and they're working on a variety of projects spanning all areas of computer science, you know, from theory, artificial Intelligence, machine learning, systems, security and MSR India also is known for its Technology for Emerging Markets (TEM) Group where we look at, you know, problems specifically affecting developing countries like India. So, essentially, Research Fellows join one of these projects and work with world class researchers on multiple phases of research, including ideations, building solutions, prototyping those solutions, deploying them in the field, working with large real industrial datasets to test out their solutions. So that experience essentially gives the perfect taste of what modern computer science research looks like for Research Fellows, and just like Shruti, most of our Research Fellows after their stint at MSR India apply for grad school and you know, go to one of the top grad schools across the world, but there are others who decide, you know, research is not for them. Many of them join Microsoft and continue to work at Microsoft in some, you know some other role. And a few of them actually, you know, have taken the entrepreneurial route. You know, we have had CEOs of many, many big companies, including Ola and some of our own projects which have been converted to startups like Digital Green and Everwell. So, you know, some of them take that route as well. But primarily it's that experience of what computer science research looks like today. I think that's the essence of what the research program offers students.

Sridhar Vedantham: Great and uh, in terms of numbers, how many Research Fellows do you think would have graduated so far from MSR India over the years?

Vivek Seshadri: Like I said, you know, in the initial years it was handful of Research Fellows that we have So overall, if you look at the numbers, it's around 150 Research Fellows in a period of 15 years. But if you look at our run rate in the most recent years, we have been graduating close to 30 Research Fellows every year, and that number is just increasing.

Sridhar Vedantham: Great, so this is another contribution of MSR to the research field, I guess.

Vivek Seshadri: Absolutely, yeah. And you know these guys hit the ground running. You know, if they do choose to do a PhD, they hit the ground running. They know how research works. They know the different phases of, you know, research projects, how to go about tackling these problems. So, in that sense I'm assuming all the advisors of past Research Fellows are extremely happy advisors- little less work for them in actually guiding their students.

Sridhar Vedantham: [Laughing] OK, I think we need to speak to some of the advisors privately to validate that.

Vivek Seshadri: Yes.

Sridhar Vedantham: So Vivek, another question for you. You know MSR India has also got this very large internship program. What's the fundamental difference between an internship at MSR and a Research Fellow position?

Vivek Seshadri: MSR used to have a very vibrant three-month summer internship program and I think over the years what we have realized is that for any stint at MSR to benefit both the student and the research projects at MSR, you need that reasonable amount of time. So, if you see even the internship program, you know we have almost faded out three-month internships. We only offer six-month internships these days. So in that sense, the main difference between the traditional internship program and the Research Fellow program is that duration. You know when a Research Fellow comes in, we know that they're going to spend at least one year with us. In most cases, they are going to spend two years with us, which means the researcher, with confidence, can set a really big goal for the Research Fellow and you know, really have that long shot instead of, a short internship where to have any meaningful output, you know, you cannot have a really large vision or a large goal. So in that sense, if you actually see the contributions that Research Fellows have made to some of our projects, they are extremely substantial, solving some fundamental problems in computer science, impacting, projects in major ways within Microsoft and also having impact on society. These contributions would not be possible in a six-month internship program.

[Music]

Sridhar Vedantham: Shruti, when you were doing your Research Fellowship at MSR India, what is the kind of work you were doing and who were your mentors out here?

Shruti Rijhwani: Right, so as I said, I was really interested in machine learning applications, more specifically as applied to natural language processing problems. So when I was a Research Fellow at MSR India, I was mentored by Monojit Choudhury and also worked closely with Kalika Bali on natural language processing problems. We were focusing on the problem of code switching, which is basically when multilingual people mix languages when they're speaking. we were basically trying to create natural language processing tools to automatically understand such code-mixed language.

Sridhar Vedantham: And how do you think that helped you? I mean you are at CMU now, which is one of the top tier PhD schools in the world. But how do you think the time that you spent here in MSR India actually helped you get to where you are?

Shruti Rijhwani: I think I credit my whole graduate school decision-making process, the application process, and even the way I do research in grad school to my experience as a Research Fellow in MSR India. Of course, the first thing was that I wasn't even sure whether I wanted to go to grad school, but after going through the Research Fellow program and with my amazing mentors and collaborators at MSR India, I took the decision to apply to grad school. And as I said, I was working on problems in multilingual natural language processing. And although I focused on code switching at MSR India, it kind of made me very interested in continuing in this field of just trying to be able to automatically process many, many languages at the same time and through my Masters and now my PhD, all of my research has been focused on doing multilingual NLP. So, in a way my experience at MSR sort of shaped my view of NLP research and taught me important research and technical skills, going all the way from like doing a literature survey to collecting data, doing experiments, and finally writing publications. I went through that process for the first time at MSR India, and it's like really helped me go through graduate school as well.

Sridhar Vedantham: Great, so it sounds like you really worked hard while you were at MSR. Did you actually have some time to do something other than work, to kind of go out, have fun and enjoy yourself?

Shruti Rijhwani: Yeah, I tried to keep a great work life balance at MSR India and in fact I think MSR and the Research Fellow program in general quite encourages that. We had a strong community of Research Fellows. We were all really good friends and it kind of makes sense because all of us were in the same exploratory phase of doing research for the first time, for most of us. So there's a lot we had in common, and all of us were good friends, enjoyed ourselves outside of work as well. So yeah, I really enjoyed my time at MSR.

Sridhar Vedantham: I believe MSR India's also got this fantastic annual offsite which is a great deal of fun.

Shruti Rijhwani: Definitely. I was a Research Fellow for one year and the offsite that year was really good fun. It's good to interact with people outside of work as well and learn about the people you collaborate with as people, not just colleagues. So I really enjoyed the MSR offsite and as Research Fellows, we also would often have our own outings like we would go out in the city and explore the city and so on. So it was – really fun. I really appreciated the community of Research Fellows that we had.

Sridhar Vedantham: Super. Vivek, a question for you. What does the Research Fellow selection process look like? How do you actually go about it, and is there any particular way in which you look at matching candidates to mentors?

Vivek Seshadri: Absolutely. I think in in many ways the Research Fellow selection process is similar to how grad school applications work. how universities select PhD students and there are multiple projects going on inside MSR India in different areas. Applicants have specific interests. You know, these days, even in the four years of undergrad that they go through, people develop interest in various areas, so they come in with their own interests. So our goal is to essentially identify the best candidates for each project. And like I mentioned, you know, it's not only the case that the number of Research Fellows slots have increased over the years from three in the first year to something close to 40 to 50 right now. It's also that the number of applicants has increased significantly. We receive close to 1000 applications and in some sense you know it's a long review process where we ensure that each applicant at least gets one pair of eyes looking at the application and then determining whether there are projects within MSR India that suit that candidate, . we come up with extensive rankings and finally researchers go through that pool of applicants that we have ranked and figure out who the best candidate for their project is.

Sridhar Vedantham: So, Shruti, if I were to ask you what the main takeaway is for you from the Research Fellow program, what would you say?

Shruti Rijhwani: Well, there are a whole bunch, I really gained so much from this program. As I already said, you know it shaped my view of research and really gave me a good footing to start my journey through Graduate School with my Master's and now my PhD. I would really recommend this program to anyone who is looking to get a little bit of experience in computer science research and if anyone wants to kind of explore this as a career option, I think this is a really good first step. Particularly because there are researchers in MSR India working on such broad fields in computer science. And just attending talks and having conversations about their research can really expand your view on computer science research and give you a really strong place to set your own career off in whatever field you're interested in. So I think that's my main takeaway. I really enjoy the community feeling in MSR India among the RFs as well as with the senior researchers in the lab. And just the fact that it helped me so much in Graduate School. I definitely recommend the program to anyone who has an inkling of interest in computer science research.

Sridhar Vedantham: That's fantastic, and I know for a fact that the people who are involved with the Research Fellow program and in general people at the lab really feel proud when, you know, people like you go through the program and then join some of the top organizations in the world.

So Vivek, how do students apply for an RF position? And what's the kind of exposure that they can look at? I mean, I know we've been speaking to Shruti, who you know, kind of was focusing on one particular area, but in a more general sense, can you talk a bit about what students can expect when they come for a Research Fellow position here?

Vivek Seshadri: Sridhar, if you look at computer science as a field in the past few years, it has become increasingly interdisciplinary. And Microsoft Research India being a flat organization, enables researchers for experts in different areas to sort of collaborate with each other and, you know, solve big problems that require such collaboration. In fact, in the past few years, if you have noticed we have had major impact in multiple problems where researchers from machine learning, systems, compilers and HCI have come together and offered big solutions to important problems. So in that sense, given where computer science is today and where it is going, I think MSR India is sort of an ideal place for new budding researchers to come and gain experience in what that interdisciplinary research looks like.

In fact, as an RF, if I mean just like Shruti mentioned, even though they are working on a specific problem in a specific area, they will often not only have the opportunity to listen to people working in other areas through talks and lectures and whatnot, they may also have to actively collaborate with other Research Fellows working in these other areas. So that experience of interdisciplinary research is sort of essential for any budding scientists in computer science. So I think that's one of the main reasons why I feel MSR India is a great place for students to come and test what research in computer science looks like today.

Sridhar Vedantham: And what's the process for somebody applying for the RF position?

Vivek Seshadri: Yeah, like I mentioned, the application process is very similar to, you know, a grad school application. You know we do require students to upload their resume and a statement of purpose. There's a portal. People can just search for the Research Fellow program at MSR. In the Internet it will lead them directly to

our page from where they can apply. There's a single deadline this time on January 15th (2021), which is right after you know grad school applications. So if students are applying to grad school, they already have all the material to apply for the Research Fellow program.

Sridhar Vedantham: Great, so what I'll do is add links to the Research Fellow program. Will attach it to the podcast so that the listeners can actually go and check it out and see what the Research Fellow program offers.

And Shruti and Vivek. Thank you so much for your time. I know that we're doing this in multiple time zones, so thank you so much for being accommodative. And it's been a great conversation.

Vivek Seshadri: Likewise, Sridhar, thanks a lot.

Shruti Rijhwani: Thanks, I had a great time.

Sridhar Vedantham: Thank you, stay safe, everybody.

[Music Ends]

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